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Conceptual design of a user-centric smart product-service system using self-organizing map

Jingchen Cong^a, Chun-Hsien Chen^b, Xuan Meng^{c,*}, Zhongxia Xiang^a, Liang Dong^a^a School of Mechanical Engineering, Tianjin University, Tianjin 300350, China^b School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore^c Beijing Jiarui Intelligent Technology Group Co., Ltd., Beijing 100089, China

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ABSTRACT

As an emerging IT-driven business paradigm, smart product-service system (Smart PSS), which offers not only the smart, connected product (SCP) but also its generated service as a solution bundle, has become a vital research topic. Many research efforts have been devoted to constructing the conceptual design framework by considering SCPs and services simultaneously. However, the following critical issues in Smart PSS conceptual design have not been well addressed: how to improve the solution of Smart PSS in the conceptual design stage to meet user emotional requirements. Aiming to fill the gap, this work proposes a conceptual design method for Smart PSS from the perspective of analyzing user-generated emotions/feelings. Specifically, the relevant traditional products are identified, and their public review data is used to analyze user emotions/feelings in user-product interaction. Interactive emotion board as a new design tool is presented to organize the user-generated emotions/feelings, associated design elements, and the potential design points of the initial solution. And the analytic hierarchy process (AHP) is utilized to evaluate the improved solution. To ensure the efficiency of the analysis process, the self-organizing map (SOM) algorithm is utilized in the process of clustering product samples and Kansei words. A case study of smart electric bicycle service system (SEBSS) design is used to demonstrate the performance of the proposed method. Based on the case study, the proposed approach appears effective in helping with Smart PSS conceptual design.

1. Introduction

With the development of SCPs (smart, connected products) and other information technologies (e.g., Internet of Things), Smart PSS (smart product-service system) is gradually flooding the market and changing people's lifestyles. Smart PSS evolves from Product-Service System (PSS) [1], which leverages product and service to work as a solution bundle to fulfill user requirements. Smart PSS was thusly proposed to combine SCP and generated services as a mixed solution [2]. Many works defined Smart PSS in various descriptions [3–5]. Among these studies, Zheng et al. [2,6] defined Smart PSS as "An IT-driven value co-creation business strategy consisting of various stakeholders as the players, intelligent systems as the infrastructure, SCPs as the media and tools, and their generated services as the key values delivered that continuously strives to meet individual customer needs in a sustainable manner". Moreover, three unique design characteristics of Smart PSS were proposed in our previous study [7]: (1) design process forms a

data-based closed-loop; (2) design means is value co-creation based on intelligent technology; and (3) design results can realize context-awareness. Some studies have endeavored to propose a novel conceptual design methodology (e.g. design entropy theory [8], function-oriented Smart PSS optimizing approach [9], context-aware concept evaluation approach [10]) to meet these characteristics. From the characteristic of the closed-loop design perspective, the existing studies emphasize completing the closed-loop design by utilizing the self-studies data collected in the usage phase, which means that Smart PSS can improve the design by using its own data after the system is produced. In fact, the closed-loop design of Smart PSS also includes improving the solution of a new Smart PSS before the usage stage by utilizing the user-generated data of existing traditional products.

Meanwhile, some products (e.g. smartphones, personal computers etc.) are now becoming mature products, and sales are not growing as before. Take smartphones and personal computers as an example. According to the statistics by the national development and reform

* Corresponding author.

E-mail address: xuan.meng@jiarui.ai (X. Meng).<https://doi.org/10.1016/j.aei.2022.101857>

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commission of China, in the first quarter of 2022, the global shipment of smartphones was 311.2 million, a year-on-year decrease of 11 %; the global shipment of personal computers was 71.3 million, a year-on-year decrease of 15.3 %. Smart PSS, which originated due to the development of digital technology, always pays more attention to technical realization. The existing studies tend to mine and meet the functional requirements of Smart PSS by inviting representative users to participate in the design process. However, with increasing competition from economic globalization, the technologies and functions are difficult to reflect the advantages of the offerings, and the product/service that can meet the users' emotional requirements is more likely to be favored by users. Manufacturing companies and development teams take unprecedented consideration on user affective experience to ensure their product and service stand out in massive convergent ones. In such context, the focus of Smart PSS development has been shifted from function achievement to user feeling optimization. It is essential to consider user-generated emotions/feelings to improve user experience and stickiness with Smart PSS [11]. However, the way to improve the solution of Smart PSS in the conceptual design stage by utilizing the feelings/emotions felt by users when they use the existing products is still lacking.

To tackle the above mentioned issues, this study aims to establish a novel Smart PSS conceptual design approach, which can be used to improve the solution of Smart PSS by considering the user-generated emotions/feelings and utilizing the online review data of existing related products. To illustrate, the solution of smart electric bicycle service system (SEBSS) is improved through this new method. Through this work, a bridge between traditional products and Smart PSS is built to improve the user experience of solutions by considering user emotions/feelings. The rest of this paper is organized as follows: [Section 2](#) mainly introduces the related research of conceptual design for PSS and Smart PSS, and affective design and Kansei mining. [Section 3](#) details the process of the proposed method. [Section 4](#) takes the SEBSS as an example to verify the feasibility and effectiveness of the proposed method. [Section 5](#) discusses the advantages of the proposed method. [Section 6](#) summarizes the research conclusions and future prospects.

2. Literature review

This section provides an overview of the relevant theories and previous studies. Firstly, the studies of conceptual design for PSS and Smart PSS are introduced. Secondly, the related research combining KE method and text mining technology is classified and summarized.

2.1. Conceptual design research from product to Smart PSS

Conceptual design is the process of completing the basic design solution by identifying the problem through abstraction, establishing structure, and selecting appropriate principles [12]. Conceptual design was initially defined as one phase in the product development process. Many existing studies proposed conceptual design methods to assist designers in obtaining product solutions accurately and efficiently. To deal with conceptual optimization, an intelligent-design method was presented with an effective functional-structure combination model that maps the product characteristics received by expert research and customer-requirements analysis of the function and structure domain [13]. To save time in design work, a systematic effect-solving approach for radical innovation of product design can be used [14]. To reduce innovation blindness, a conceptual design method driven by product genes was advanced, and a functional expansion model was established based on analyzing the conceptual design process [15]. To decrease designers' time in capturing design information from a large volume of unstructured textual data, a function-structure concept network construction approach integrates sentence parsing and word/phrase extraction to combine functional and structural information [16]. Meanwhile, some studies focused on solving the problem designers may

face in the product conceptual design process, e.g. the way to synthesize multi-disciplinary principles together to get a solution [17]. Furthermore, users not only focus on buying the right products but also pursue service experiences now [18]. The service blueprint was combined with other methods (e.g. TRIZ, Quality Function Deployment) to complete service design [18,19]. Moreover, a service design approach integrating Kansei Engineering and a data mining technique was proposed to incorporate the customers' feelings into the service design [20]. However, the conceptual design methods of products or services tend to conceive products or services separately.

PSS, which integrated products and services as a solution bundle, needs to be conceived as a whole. The PSS layer method was proposed to analyze and synthesize PSS ideas and concepts in the early development phases [21]. The concept map was presented to structure and evaluate the artifacts in the early phases of the design process before proceeding to a detailed design [22]. Some existing studies focused on dealing with the relationship between product and service. An integrative PSS design method was presented from a human-oriented perspective [23], and a four-phases methodology for planning the module portfolio of the PSS service solution layer was explained to get maximum customer satisfaction [24]. Furthermore, some studies focused on transforming customer requirements into PSS solutions and considering value-added factors (e.g. sustainability [25]). A customer-driven practical approach can be used to analyze the correlation between customer expectations, the PSS components, and the PSS modules [26]. A systematic PSS approach was developed to determine the optimal fulfilment levels of product-related and service-related engineering characteristics by considering the requirements of customers and manufacturers [27]. Meanwhile, a PSS development method can be utilized to enhance sustainability through the simulation of life-cycle scenarios [28].

With the development of Smart PSS paradigm, some works began to discuss the Smart PSS conceptual design methods based on the relevant studies of PSS. User is the centre of product and service development activities, and Smart PSS aims to cater to diverse user requirements [29]. Hence, some studies proposed novel design approaches for Smart PSS from the user-oriented perspective, e.g. the new user-centric Smart PSS development method based on the five layers of user experience design [29], the technique based on TRIZ functional model and AHP [30], the innovative value propositions identification approach [31], the novel framework to capture and evaluate customer needs, co-creative value propositions and system requirements [32], the user-centric Smart PSS design method by utilizing the virtual reality system-generated data [33], the new Smart PSS design method by using user-generated data [34], and so on. Features of Smart PSS design methods are presented in [Table 1](#).

Overall, none of the above studies has offered an in-depth method for improving Smart PSS solutions concerning user-generated emotions/feelings. Although some deal with the users' emotional needs by introducing the user experience method in service design, few studies leverage the user comment on similar products to obtain potential design points during the user experience process. Therefore, this paper aims to develop a conceptual design method that can be used in optimizing the initial solution of Smart PSS by using the massive existing online user review data.

2.2. Text mining in Kansei Engineering

Affective design is introduced to deal with user emotions/feelings in this paper. In the context of affective design, the term "emotional" is often used as equivalent to the term "affective" [35]. Some methods can be used to integrate affective aspects into product development, e.g. Semantic Differential Methods, Conjoint Analysis, Semantic Description of Environments, Quality Function Deployment, and Kansei Engineering [36]. Among these methods, Kansei Engineering, proposed by Nagamachi [37], can be used to satisfy users' emotional perceptions of services and products [38]. Kansei is an internal concept with three basic

Table 1
Features of Smart PSS design methods.

Ref.	Concern	Foundation	Objective	Outcome
(Chang et al. 2019) [29]	Multimodal user experience	Service dominant logic	Outline the conceptual framework of user-centric SPSS and contribute a user-centric SPSS development approach	User-centric SPSS solution (service logic, provider integration network, and hardware design)
(Cong, Chen, and Zheng 2020) [8]	User-generated information	Information theory and adaptable design	Take the process of information prediction, summarization, conversion and updating into an overall consideration	Information conversion map (with information which has solution mapping)
(Wu et al. 2020) [30]	Function	TRIZ functional model and AHP	Construct conceptual design scheme and divide modules from the functional perspective	Functional model diagram and selected scheme
(Chen et al. 2020) [31]	Product, user and industry perspectives	Innovative value propositions (IVPs)	Identify and evaluate IVPs in the context of smart PSS	Five types of IVPs in five value scopes and their corresponding importance
(Liu et al. 2020) [32]	Conceptual factors (i.e. CNs, CVPs and SRs)	Conceptual factors for PSS design	Construct framework and method to capture and evaluate the conceptual factors for smart PSS.	Sequence diagram for interaction procedures and prioritization of conceptual factors
(Wu et al. 2021) [9]	Function	Five-dimensional digital twin framework (i.e. physical entity, virtual entity, service module, DT data module, and connection module)	Improve Smart PSS solutions at the conceptual design stage based on the framework of DT	Improved Smart PSS solution
(Yang et al. 2021) [11]	User emotion	AHP and Fuzzy comprehensive evaluation (FCE)	Construct a user requirement framework and quantitatively distinguish the influence of each requirement	User requirement framework and the influence of each requirement
(Bu et al. 2021) [33]	User-generated data and VR system-generated data	VR technology	Outline a conceptual approach for user-centric development of value-added Smart PSSs leveraging users-generated data and VR system-generated data	Overall architecture consisting of three layers (i.e. VR platform development layer, VR data layer, and VR value-added service layer)
(Chiu et al. 2021) [34]	User-generated data	Machine learning and NLP	Analyze user-provided data and offer customers with personalized solutions based on NLP and deep learning	Personalized recommendation solution

pillars of taste/sentiment, feeling and emotion [36]. Often Kansei words are adjectives to present users' emotions and feelings, but other grammatical forms are possible [36]. A general framework of KE includes five main phases [39]: (1) choose the design domain, (2) span the semantic space and property space, (3) synthesize the semantic space and property space, (4) test the validity, and (5) build models. Although originally, the KE-related studies focused on physical products, KE is also suggested to be a creative and practical way to improve the existing services and innovate new ones [39,40]. Related researches apply KE to the design of the smart home service [41], travel service [42], short distance delivery service [20,43–45], international express service [46], logistics service [38,47–49], hotel service [50,51], airport services [52], smart government service [53] and so on. These studies show the possibility of KE applying for both products and services. What's more, the extended KE approach for the design of PSSs developed by [54] and applied to designing bus bodies and transportation services can show the value of KE to PSS. In this study, the idea of KE will be further integrated into the conceptual design process of Smart PSS.

Meanwhile, text mining is introduced to deal with user reviews online. Text mining refers to a technology that obtains valuable information and knowledge from unstructured, semi-structured, or weakly structured text data [55], especially in the internet context. Relevant studies combining KE method and text mining technology are divided into three categories: 1) First kind of work is conducted in specific industries and does not emphasize the innovation of methods but focuses on the innovation of application fields. Through the combination of KE and text mining, instructional advice is given to the development work in related industries, both product and service industry [38,42,51,56]. 2) Second kind of work aims to improve the KE process or expand the KE structure. Through the application of text mining technology, KE can better deal with massive online user-generated content to make the corresponding product design process more efficient. Related studies raised KE methods by combining text mining to enhance the ability to capture user-generated information in online reviews, mapping customer emotional needs with design elements and constructing the Kansei evaluation model [57–64]. The main contribution is to improve the effectiveness and efficiency and increase the automaticity of KE

methods. 3) Third kind of work emphasizes the proposal of innovative methods by applying KE and text mining to solve specific problems. Jin et al. [65] analyzed online reviews at a fine-grained level and extracted aspects of product features and detailed reasons of consumers to inform designers regarding what leads to unsatisfied opinions. Zhang et al. [66] proposed an opinion mining extraction algorithm to jointly discover the main opinion mining elements. Jin et al. [67] illustrated a framework to select pairs of comparative sentences with specific product features. Liu et al. [68] proposed a method based on the sentiment analysis and the intuitionistic fuzzy set theory to rank the products through online reviews. Othman et al. [69] proposed a conversation-based method with a reply tree to effectively extract the product features involved in the conversational element of Twitter. Jin et al. [70] proposed a Kano-based method to analyze the importance of product features on customer satisfaction and instruct innovative product design. Lian et al. [71] proposed a novel affective-blue design approach by integrating KE, text mining and machine learning to support sustainable, healthy, but cheap offerings.

The first kind of work lacks universal applicability due to the limitations of its application field and cannot provide sufficient inspiration for other studies. The second and third kind of work emphasizes the improvement and expansion of the KE method and proves the potential of the combination of KE and text mining in assisting the design process. Nevertheless, these methods have not provided a systematic and specific process for Smart PSS, especially the way to utilize the review text of traditional products in improving the Smart PSS solution, which is the topic of this research. It should be noted that Kim et al. [72] developed a methodology to reflect users' implicit requirements from users' affective variables, which are extracted from online reviews and classified by using the SOM clustering method. This study is the basis of our proposed method, but there are some differences between these two approaches: (1) Different goals. The goal of the existing study is to develop an affective variable extraction methodology that can reflect users' implicit needs effectively and efficiently, but our study is to propose a conceptual design improvement method based on affective extraction technology. (2) Different design objects. The design object of the existing method is the product, but the proposed method is Smart PSS. (3) Different

processes. The existing method used text mining techniques and the SOM to gather and analyze users' affective experiences, but the proposed method used these technologies to select reference products and gather users' emotions to improve the design solutions. The proposed method of this paper is aimed at product-oriented Smart PSS, which has similar traditional products with a large amount of review data.

3. Method

The flowchart of the proposed method is depicted in Fig. 1. Three key steps can be classified as 1) identifying the reference products and collecting their online review data, 2) making an interactive emotion board through online review of reference products, and 3) improving the initial solution based on user-generated emotions/feelings. It should be emphasized that before this improvement process starts, an initial

solution of Smart PSS with SCPs has been output by designers.

3.1. Stage 1: Identifying the reference products and collecting their online review data

As shown in Fig. 2, identifying the reference products and collecting their online review data can be divided into three steps, which are elaborated on below.

1) *Stage 1- Step 1: Identifying the traditional product category.* At first, select one usage scene of Smart PSS as the specific usage scene to be considered in this design work. All components of the SCP(s) in the initial solution under the specific usage scene should be listed. SCP in Smart PSS is composed of conventional physical components, smart components, and connectivity components [73] to realize the functions of monitoring, control, and optimization. The smart components consist

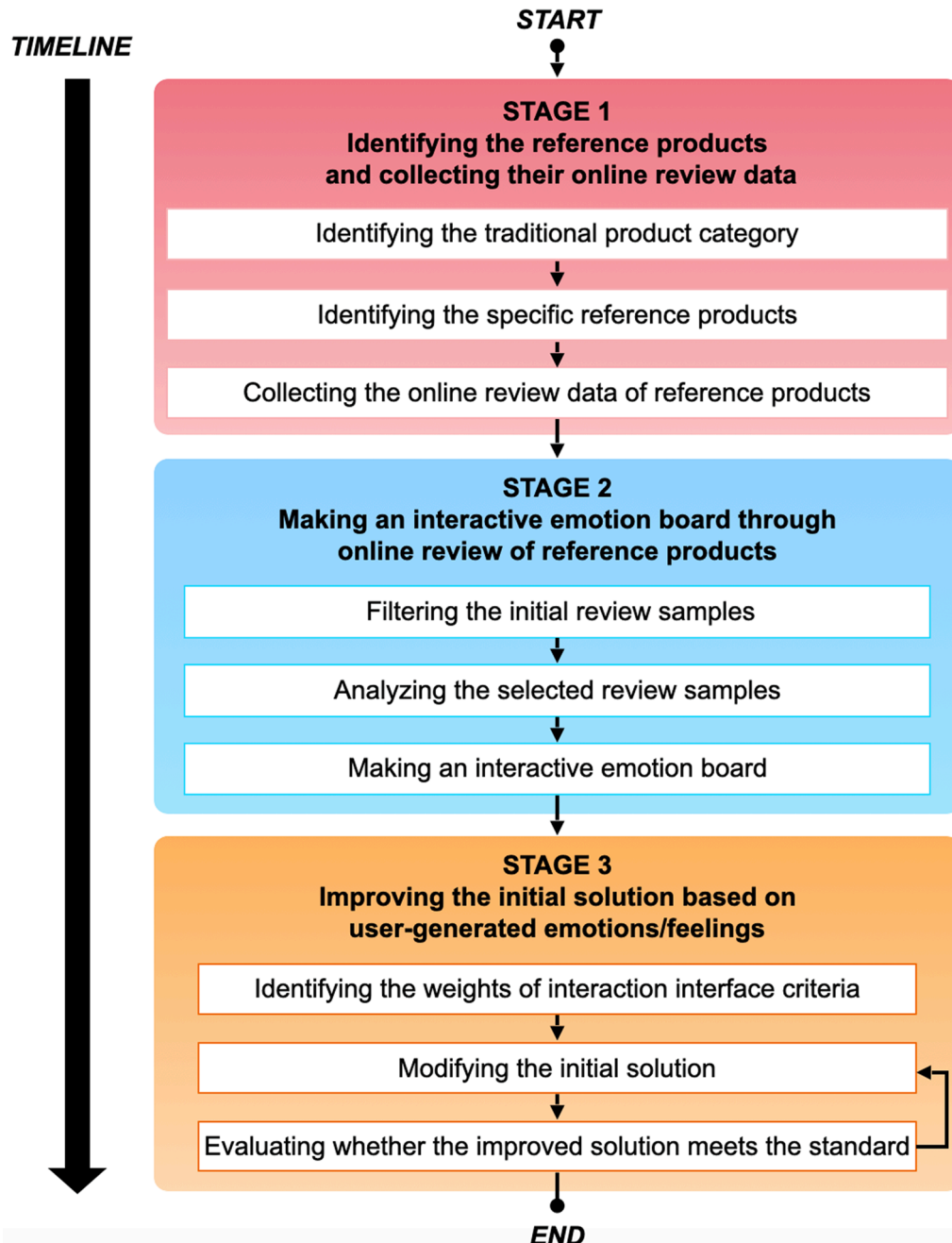


Fig. 1. Flowchart of the proposed method.

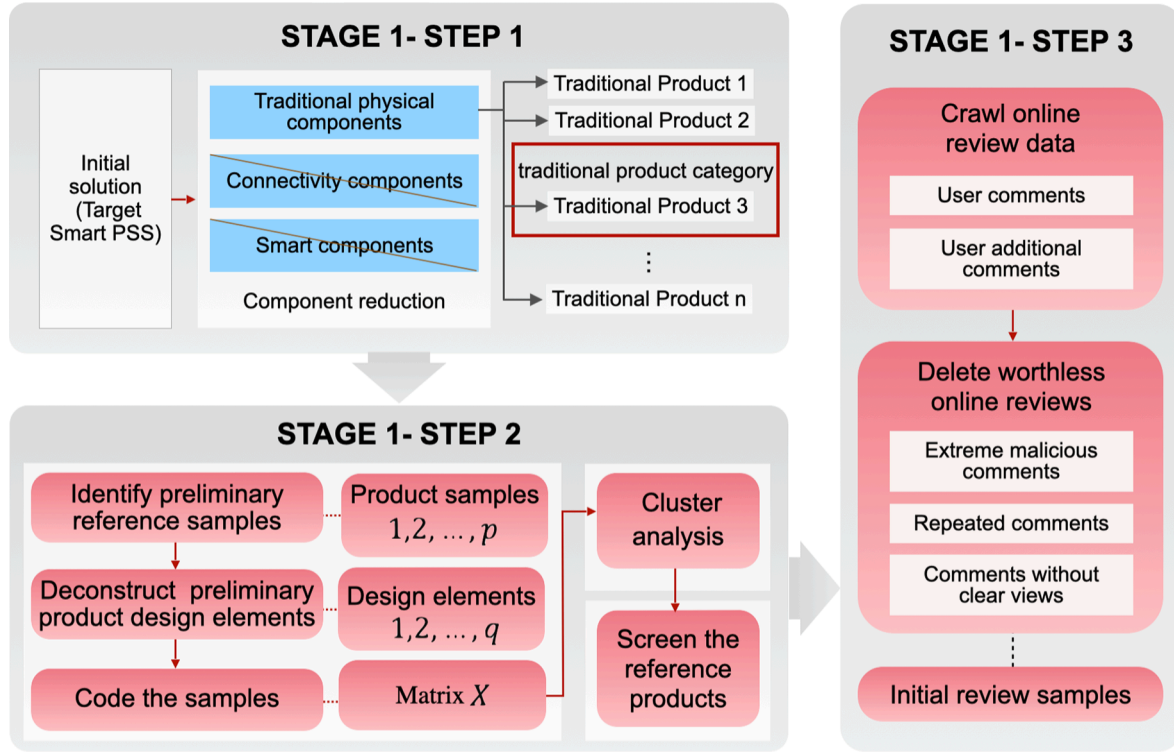


Fig. 2. Process of stage 1.

of microprocessors, sensors, controls, data storage, software, and embedded operating system [74]. Meanwhile, the connectivity components consist of the product's antennae, ports, and protocols enabling wireless or wired connections [74]. Then, smart and connectivity components should be filtered out from all the components. Because Smart PSS sometimes includes some physical products to work together, the remaining components may comprise one or more traditional products. The designers should select the product which interacts with users most frequently under the specific usage scene and has the greatest impact on user experience as the objects to improve. The selected product can be regarded as the traditional productcategory to be referenced. Meanwhile, its key user-product interaction interfaces can be summarized by designers under the specific usage scene.

2) Stage 1- Step 2: Identifying the specific reference products. The identified traditional product category can be searched on an e-commerce website. Then, the products with the highest sales can be taken as the preliminary reference samples, marked as 1, 2, ..., p . The number of the preliminary reference samples is determined by the computing conditions of the development team. Meanwhile, the product design elements that may affect the user experience during the interaction process are sorted as preliminary product design elements marked as 1, 2, ..., q . Moreover, the developers can collect the samples' introduction and description on the website through the crawler software and then conduct binary coding according to the performance of each product sample on each design element. Each product is represented as a q -dimensional vector. The i -th number in the vector represents whether the product contains the i -th design element, 1 represents including, and 0 represents not including. Then a $p \times q$ dimensional matrix is formed, recorded as matrix X .

Then, cluster analysis is carried out on matrix X . K-means, as a classical algorithm for clustering, is often used in clustering analysis as an iterative unsupervised learning algorithm. Though K-means features a simple structure and fast running speed, it is easy to cause non-convergence or local optimization, and the number of clusters and initial cluster center must be given first [75,76]. By comparison, SOM is effective for feature extraction, dimension reduction, and data

visualization, with the ability to perform clustering while preserving topology. It can transform complex, nonlinear statistical relations between high-dimensional data into simple geometric relations at a low-dimensional display. The result of the SOM process is easy to read and understand, which makes the proposed method easy to be executed [72,77]. Consequently, SOM, as a visual unsupervised learning algorithm that can reduce the dimension of data and cluster [78], is conducted to complete the clustering analysis in the proposed method. It is utilized to cluster product samples and online user reviews to generate a visualized map, which can assist designers in obtaining valuable information more effectively. The structure of SOM is shown in Fig. 3 [72]. For the input Matrix X , the Euclidean distance between X and weight W is estimated to identify the node with the minimum distance value, and this identified node is the winner node [72]:

$$D_{ij} = \sum_{i=1}^n (w_{ij} - x_i)^2 \quad (1)$$

where n is the length of the input vector, w_{ij} is the value in column i of row j in the weight table, and x_i is the i -th value of the input vector [72]. Then the winner node weights are revised:

$$w_{ij}(\text{new}) = w_{ij}(\text{old}) + \alpha(t)(x_i - w_{ij}(\text{old})) \quad (2)$$

where $w_{ij}(\text{new})$ is the new weight vector, $w_{ij}(\text{old})$ is the old weight vector, and $\alpha(t)$ is a learning restraint due to iteration progress [72]. Each instance of input data can be assigned to an individual cluster by utilizing the latest weights of the feature map.

After the SOM cluster analysis of matrix X , clusters are formed in a SOM plane, and each instance of input data is assigned to an individual cluster. For a more intuitive analysis, the sample number can be replaced with the picture of each product sample to obtain a new overall figure. Similar products will be projected into a lattice, and all the products in one (or adjacent) lattice(s) will be regarded as one product group. In the clustering result, multiple product groups can be formed. The developer needs to select the product group (one or more) with the highest similarity to the SCP sketch/CAD model of an initial solution of

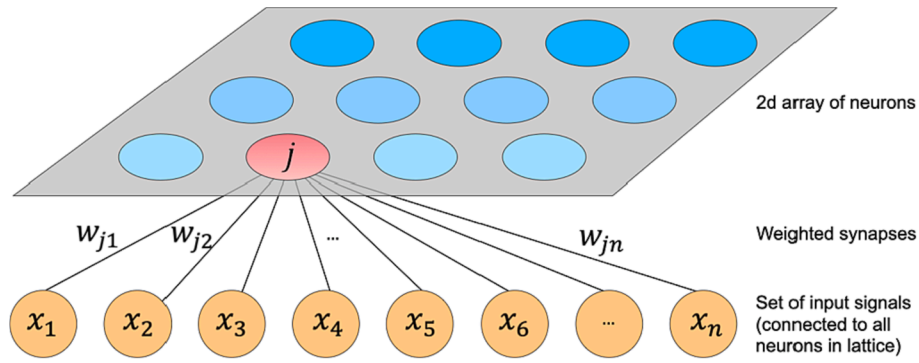


Fig. 3. Overall structure of SOM [72].

Smart PSS. The products in the selected group are the specific reference products for Smart PSS.

3) Stage 1- Step 3: Collecting the online review data of reference products. After selecting the appropriate reference products, their online review data can be crawled from the e-commerce website. The crawled data includes user comments and user additional comments. Then, the worthless online comments (i.e. extremely malicious comments, repeated comments, and comments without clear viewpoints) need to be eliminated, in order to obtain the initial review samples.

3.2. Stage 2: Making an interactive emotion board through online review of reference products

As shown in Fig. 4, making an interactive emotion board through online review of reference products can be divided into three steps, which are elaborated below.

1) Stage 2- Step 1: Filtering the initial review samples. Word segmentation is performed to separate the initial review samples into the form of punctuations and words. And data cleaning is carried out to delete the punctuations and the stop words so as to form an initial vocabulary. Word frequency analysis, as the essential means of text mining, can be used to count the frequency of words in the initial vocabulary in all initial review samples. If the word frequency of a word is less than

2, this word is not the consensus of users on the product, which can be deleted. After that, experts in the specific field can be invited to assist the developing team in completing the vocabulary screening to form a specific product vocabulary, which contains Kansei words and product design words. It should be noted that the product design words here are vital design elements affecting user experience. Furthermore, the initial review samples should be screened by removing the samples which did not mention any words in the specific design vocabulary. After the screening, the selected review samples, which are recorded as $1, 2, \dots, l$, can be used in the next step.

2) Stage 2- Step 2: Analyzing the selected review samples. At first, the Kansei words in the specific design vocabulary, which cannot be further associated with the specific usage scenes, should be deleted. After the deletion, the remaining Kansei words here are called target Kansei words, which all belong to one dimension-target Kansei element. Meanwhile, the key user-product interaction interfaces of the initial solution can be recorded as $1, 2, \dots, n$. Each interface contains many target product design words selected from the product design words in the specific product vocabulary, and these words are strongly correlated to the interaction interfaces. In order to cluster the selected review samples, text vectorization should be conducted to convert the unstructured text into a group of vectors. Bag-of-words model is a text vectorization method that takes the word as the basic processing unit [79]. One review

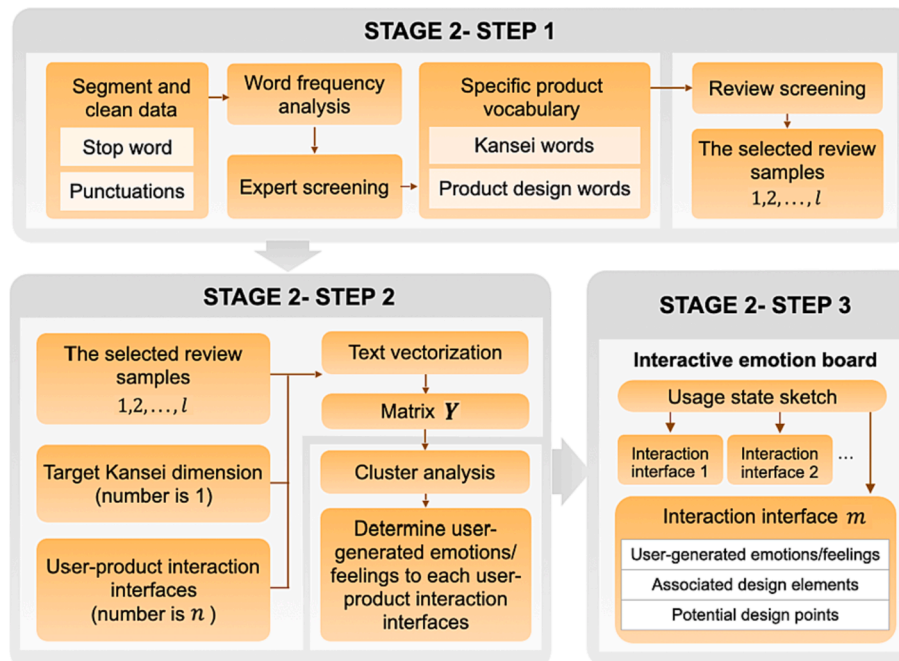


Fig. 4. Process of stage 2.

sample can be converted into one vector with $n+1$ elements. For each element, if this sample mentions the target Kansei words or the target product design words, the value of this element in the vector is 1. If this sample is not mentioned, the value of this element in the vector is 0. All selected review samples (number is l) can be vectorized into an $l \times (n+1)$ dimensional vector, recorded as matrix Y . Cluster analysis of matrix Y should be carried out. The clusters that can obviously show the relationship between target design words and target Kansei words are valid clusters.

More often, this kind of correspondence is a complex relationship, not necessarily a one-to-one or many-to-many relationship. The clusters, which reviews seldom mention the target product design words and target Kansei words, is considered as invalid cluster. Meanwhile, the clusters associated with too many interaction interfaces are also invalid clusters since they cannot make people focus on certain problems. Then word frequency analysis of target Kansei words and target design words in the comments of each valid cluster should be conducted. The valid clusters can be connected with certain corresponding interaction interfaces by analyzing the top high-frequency words in the comments of each valid cluster. The developer selects the appropriate words from the high-frequency Kansei words in each valid cluster, and records them as the user-generated emotions/feelings for the corresponding interaction interface. The same operation is also performed on design words, and high-frequency design words affecting the user emotions/feelings are recorded as associated design elements. Meanwhile, the associated design elements can also be analyzed by designers from specific interface's comments which have mentioned the selected emotional words. After analyzing each valid cluster, the user-generated emotions/feelings, associated design elements are assigned to the specific interaction interface, with a small amount of related review data.

3) Stage 2- Step 3: Making an interactive emotion board. The results of the previous step can be filled into the interactive emotion board. The

interactive emotion board is a new figure which added a table next to the interaction interface on the usage state sketch. The table includes the user-product interaction interface, the user-generated emotions, associated design elements and potential design points. The potential design points here are for the initial solution of Smart PSS, not for the reference products. They should be obtained from the analysis by combining the characteristics of Smart PSS based on the user-generated emotions and associated design elements. These potential design points can be obtained by the designer evaluating the performance of the initial solution on the associated design elements. The developing team can also summarize the potential points by conceiving that: if the smart/connectivity components which have been deleted in Stage 1-Step 1 are added to the corresponding interfaces in the traditional product, the points that should be noticed to get positive users' emotions/feelings are the potential design points.

3.3. Stage 3: Improving the initial solution based on user-generated emotions/feelings

As shown in Fig. 5, improving the initial solution based on user-generated emotions/feelings can be divided into three steps, which are elaborated on below.

1) Stage 3- Step 1: Identifying the weights of interaction interface criteria. In this step, the performances of interaction interfaces are taken as criteria, and recorded as $ic_1, ic_2, \dots, ic_n$. If the design elements of an interactive interface cannot be improved and evaluated in the conceptual design stage, this interface does not need to be taken as a criterion. AHP[80] can be utilized to get the importance of these criteria when evaluating the solutions. Moreover, $sic_{pq}(p, q = 1, 2, \dots, n)$ is used to record the scale values given by 3–5 design experts, which represents the comparison result between ic_p and ic_q . As shown in Table 2 [81], the rules can be used to give the scale values.

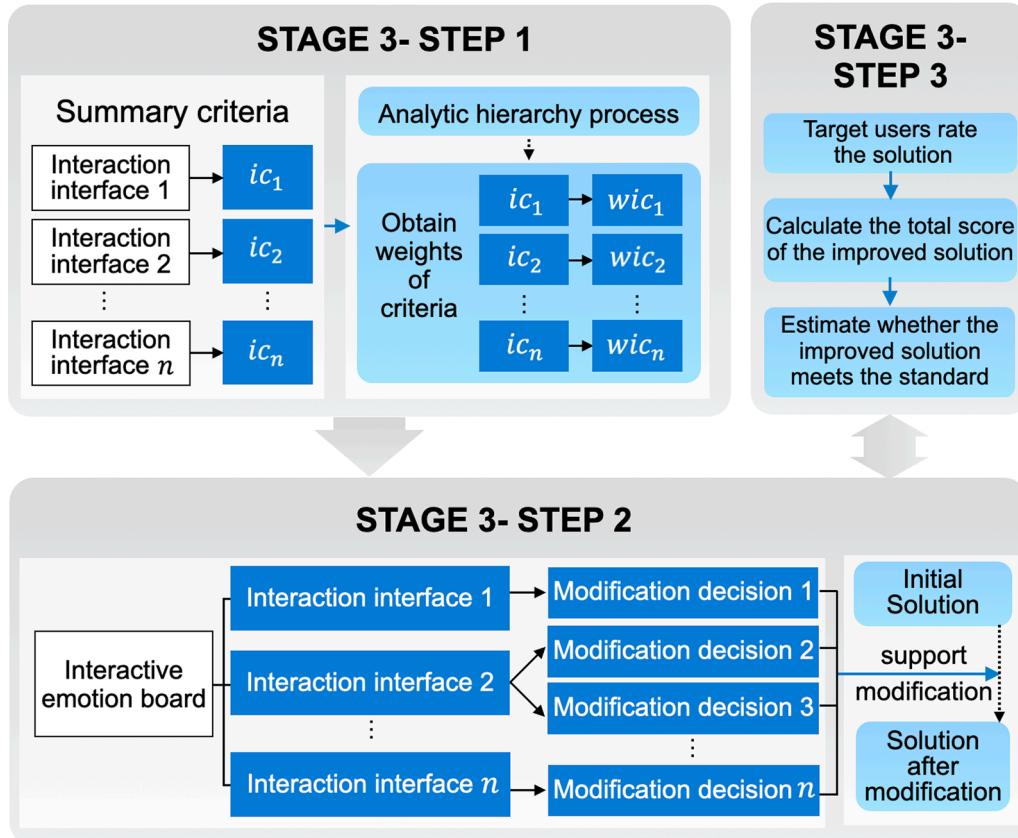


Fig. 5. Process of stage 3.

Table 2
Scale for the comparison between ic_p and ic_q . [81].

Scale	Definition
1	equal importance between ic_p and ic_q
3	the importance of ic_p is slightly higher than the importance of ic_q
5	the importance of ic_p is higher than the importance of ic_q
7	the importance of ic_p is strongly higher than the importance of ic_q
9	the importance of ic_p is extremely higher than the importance of ic_q
2,4,6,8	Intermediate values between the two adjacent judgments.
Reciprocals	If ic_p has one of the above numbers assigned to it when compared with ic_q , then ic_q has the reciprocal value when compared with ic_p .

The judgment matrix SIC of each experts can be obtained [81]:

$$SIC = (sic_{pq})_{n \times n} = \begin{bmatrix} sic_{11} & sic_{12} & \cdots & sic_{1n} \\ sic_{21} & sic_{22} & \cdots & sic_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ sic_{n1} & sic_{n2} & \cdots & sic_{nn} \end{bmatrix} \quad (3)$$

where n represents the number of criteria. Meanwhile, calculate the consistency index of the judgment matrix:

$$CI = \frac{\lambda_{max} - i}{i - 1} \quad (4)$$

where λ_{max} refers to the maximum eigenvalue, and i refers to the order of the matrix [81]. The random consistency indicator RI is obtained according to the random consistency indicator table. Calculate the consistency ratio value through:

$$CR = \frac{CI}{RI} \quad (5)$$

If the CR value is greater than or equal to 0.1, it means that the judgment matrix given by this expert has not passed the consistency checking. It is necessary to delete the score of this expert, or ask him to check and modify his score. Meanwhile, the CR value of fewer than 0.1 means the judgment matrix can pass the consistency checking. Then, obtain the column normalized matrix $NSIC$ by dividing every number in the judgment matrix by its column sum. The average value of each row of matrix $NSIC$ is the weight of the selected criteria of one expert. The average value of all obtained weights on one criterion is the final weight value of this criterion [81], and the weights of interaction interface criteria can be recorded as $wic_1, wic_2, \dots, wic_n$.

2) Stage 3- Step 2: Modifying the initial solution. According to the contents of each interaction interface in the interactive emotion board summarized in stage 2 (especially the potential design point), the modification decisions on each interaction interface should be given. Then, the initial solution should be modified according to these decisions.

3) Stage 3- Step 3: Evaluating whether the improved solution meets the standard. The solution after improvement should be scored by target users through 7 points Likert scale. Users should score according to their real feelings about the solutions, point 7 represents extreme satisfaction, and point 1 represents severely dissatisfied. The average satisfaction score under each criterion can be calculated, recorded as $A_1, A_2, \dots, A_n$. The final total score of the improved solution is recorded as S , which is calculated through:

$$S = A_1 \times wic_1 + A_2 \times wic_2 + \dots + A_n \times wic_n \quad (6)$$

Finally, estimate whether the total score of the improved solution meets the standard. The design team selects the standards here according to the specific project: 1) If the design team wants to focus on eliminating the users' dissatisfaction, the value of S should be greater than 4 (representing the users' neutral attitude towards the solution) or 4.5 (representing the users' mild satisfaction with the solution). 2) If the design team wants to focus on improving the user experience, the value of S should be greater than 5 (representing that the user is satisfied with

the solution). 3) If the design team is committed to pursuing perfection, the value of S should be greater than 6 (representing that the user is very satisfied with the solution). Then, when the value of S meets the standard, the solution can be used in the subsequent embodiment and detail design stage. When the value of S does not meet the standard, the design team needs to conduct offline interviews with the users with low satisfaction scores to further modify the design solution until it meets the standard. In addition, designers are suggested to use this method to analyze the user-product interaction interfaces in the specific usage scene. However, for some special cases (e.g. designers want to obtain more information rapidly, or analysis of the usage scene one by one cannot meet the development needs), designers also can analyze the touch points in all scenes. The analysis method is the same as the above method, except that the user-product interaction interfaces should be replaced by the touch points that users contact with the product/service. In stage 1, other steps remain unchanged, except that designers should list all touch points in all scenes instead of interaction interfaces in the specific usage scene. Moreover, the interactive emotion board summarized in stage 2 includes the touch point, the user-generated emotions, associated design elements to indicate the potential design points, and the usage state sketch that can be omitted. In stage 3, the interaction interface criteria can be replaced with touch point criteria. It should be noted that this research recommends that the designers analyze individual scenes one by one because considering all scenes at the same time needs to process large amounts of data, which may influence the accuracy and efficiency of the analysis.

4. Case study

The empirical case study is a kind of effective model to verify the presented methods [82,83,84]. Therefore, this section provides a case study of SEBSS to verify the effectiveness of the proposed conceptual design method. The conceptual design of SEBSS aims to meet the various needs in the process of using smart electric bicycles. As a system, SEBSS mainly comprises a smart electric bicycle, a smart charging pile and an application on smartphones. The user experience map of SEBSS is depicted in Fig. 6, which describes five usage scenes of SEBSS and their corresponding user sub-goals and behaviors. The overall service situation of SEBSS can be learned from the user experience map. Moreover, the Scene 3-riding scene is used in this case.

1) Stage 1: Identifying the reference products and collecting their online review data. In the riding scene, SCP involved is the smart electric bicycle. The initial solution of the smart electric bicycle is shown on the left of Fig. 7. Firstly, in order to identify the traditional product category that can be referenced, smart and connectivity components were deleted, and the traditional product composed of the remaining components was the electric bicycle, which is taken as the referenced traditional product category of SEBSS in the riding scene. And key user-product interaction interfaces under the riding scene are summarized into four interfaces (i.e. hand interaction interface, foot interaction interface, sitting interaction interface, and riding interaction interface), which are depicted on the right of Fig. 7.

Secondly, as the identified product category, the electric bicycles were searched on the e-commerce website. 20 products with the highest sales were selected as the preliminary reference samples. Meanwhile, 22 product design elements, which belonged to 8 items (i.e. materials, degree of bicycle body shell protection, riding mode, bicycle length, load, rim size, gear, and the number of seats), were identified as preliminary product design elements. Each sample was binarized according to its performance on the elements. Then, the matrix X_E (20×22 dimensional vector) was obtained. SOM Toolbox 2.1 based on the MATLAB platform, as a common tool to realize the SOM algorithm, was used on the cluster analysis of the matrix X_E . The selected parameters were listed below: the shape of lattices was hexa, the number of lattices was 8×8 , the pattern of neighborhood function was gaussian, the neighborhood radius of rough training was [3,1], the neighborhood

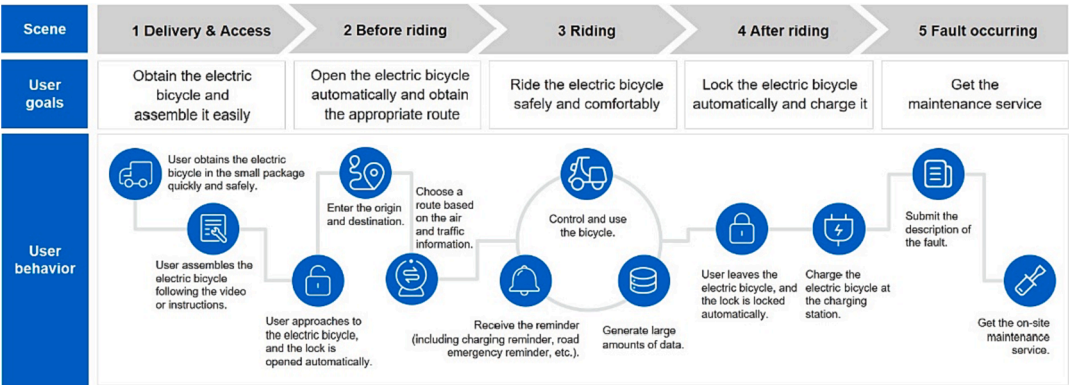


Fig. 6. User experience map of SEBSS.

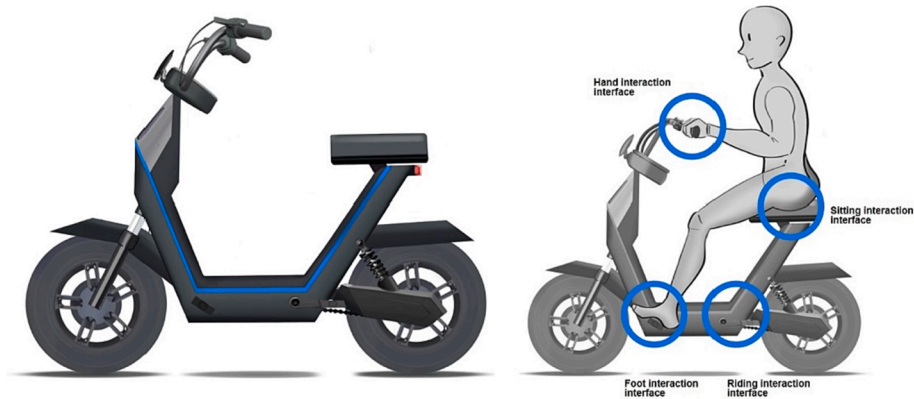


Fig. 7. Initial solution of smart electric bicycle and its interaction interfaces.

radius of fine training was [1,1], the number of training cycles of rough training was 30, and the number of training cycles for fine training was 20. SOM cluster analysis result of 20 samples was shown in Fig. 8, and the pictures of each product sample were added to the SOM plane to obtain Fig. 9. The distances between the lattices that the samples projected on the plane reflected the similarity of the samples. Four product

groups were divided, including group 1 (incl. samples No. 1, 8, 10, and 11), group 2 (incl. samples No. 3, 4, and 13), group 3 (incl. samples No. 5 and 20), and group 4 (incl. samples No. 14, 18, 12, and 19). The products of Group 1, which were most similar to the SCP of the initial conceptual design solution, were selected as the specific reference products for SEBSS. Finally, online comments of these 4 reference products on the

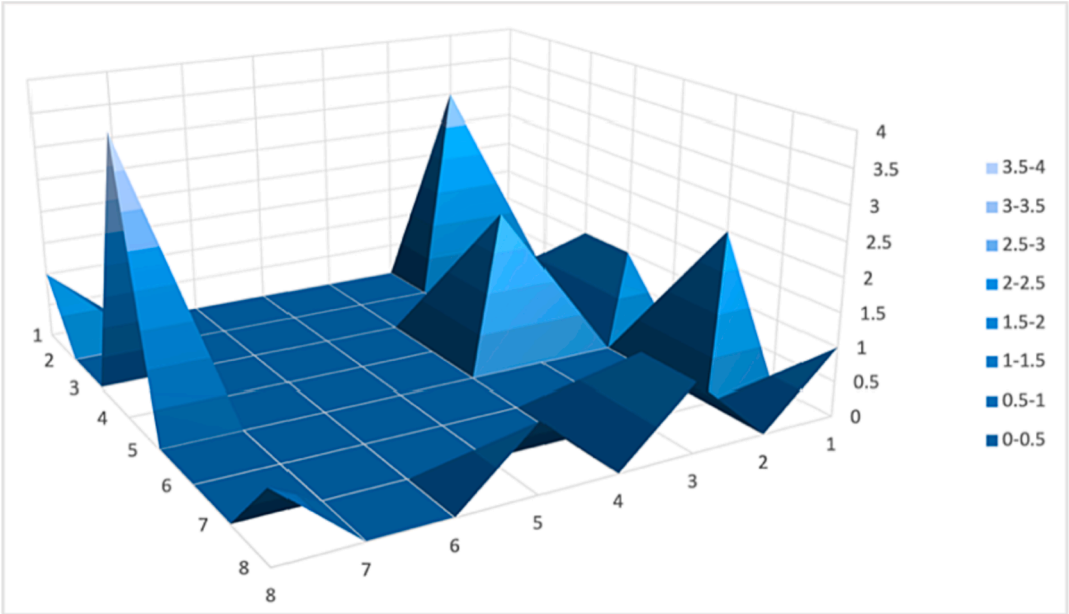


Fig. 8. SOM cluster analysis result of preliminary reference samples.

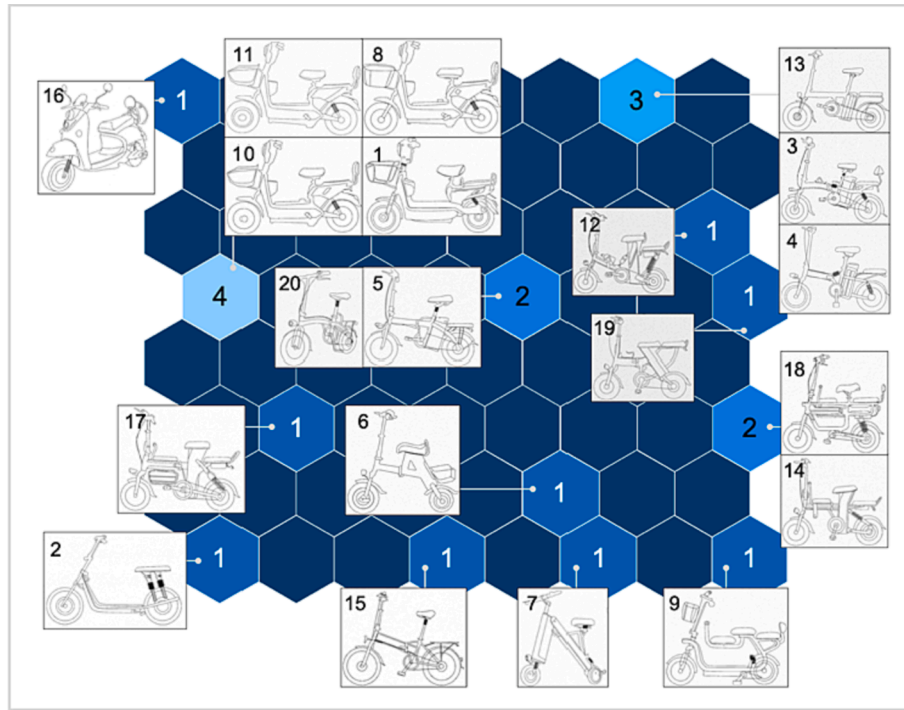


Fig. 9. SOM plane with product pictures.

Tmall e-commerce website were crawled. After cleaning the worthless comments (e.g., “this user has no comments”), 4080 items were selected.

2) Stage 2: Making an interactive emotion board through online review of reference products. Firstly, the Chinese word segmentation tool (i.e. *Jieba*) is taken to segment the selected comments. And then, the punctuations and stop words in Baidu stop word list, Harbin Institute of technology stop word list and Sichuan University intelligent laboratory stop word list [85] were deleted to form an initial vocabulary (including 6308 words). Word frequency analysis was conducted on these words through the script written in Python, and the words, which word frequency was 1, were deleted. Three enterprise experts who have been engaged in electric bicycle design for many years were invited to screen the vocabulary jointly, and 678 words were selected to form a specific product vocabulary for SEBSS. Furthermore, the review samples without any words of the SEBSS specific product vocabulary were removed, with 3964 valuable review samples remained.

Secondly, the target Kansei words and product design words were selected. Then review samples were converted into matrix Z_F (3964×5 dimensional vector), and SOM cluster analysis is performed on it. The selected parameters were listed below: the shape of lattices was hexa, the number of lattices was 7×7 the pattern of neighborhood function was gaussian, the neighborhood radius of rough training was [3,1], the neighborhood radius of fine training was [1,1], the number of training cycles of rough training was 50, and the number of training cycles for fine training was 30.

The clustering result was shown in Fig. 10, and it can be found that six clusters were generated. The reviews in cluster 1 seldom mentioned the targeted design words and targeted Kansei words, which is considered an invalid cluster. Word frequency analysis of target words in the comments of clusters 2–6 was conducted, as shown in Table 3. Cluster 2 was classified as belonging to the riding interaction interface, cluster 3 belonged to the hand interaction interface, clusters 4 and 5 belonged to the foot interaction interface, and cluster 6 belonged to the sitting interaction interface. Then, the user-generated emotions of each interaction interface were determined from the high-frequency Kansei words under each interaction interface.

Thirdly, the interactive emotion board of the riding scene was

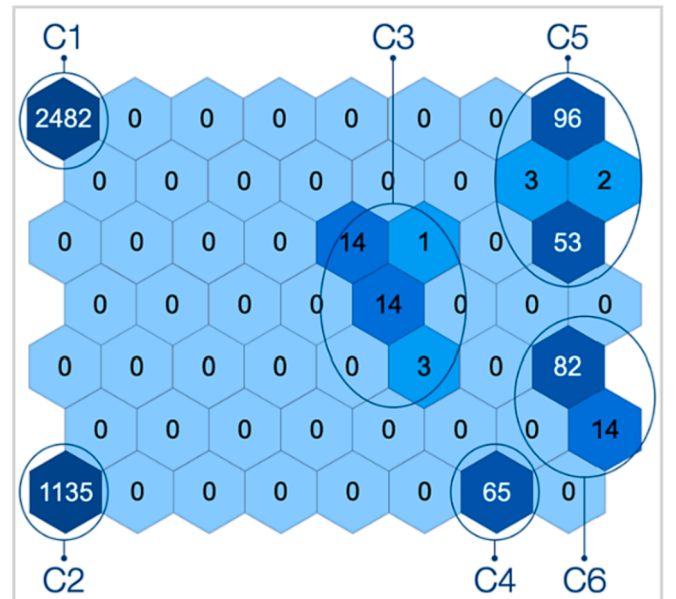


Fig. 10. SOM cluster analysis result of the riding scene.

completed, as shown in Fig. 11. User-generated emotions and the associated design elements were selected from the high-frequency Kansei words and their comments under each interaction interface. The potential design points were also listed on the board, which need to be improved in the conceptual design solution of SEBSS.

3) Stage 3: Improving the initial solution based on user-generated emotions/feelings. Three criteria (i.e. ic_1 , ic_2 , ic_3) correspond to three interactive interfaces (i.e. hand interaction interface, sitting interaction interface, foot interaction interface). The solution to the riding interaction interface can not be improved and evaluated in the conceptual design stage, so the riding interaction interface is not used as the criteria to compare the solutions. Then, 5 experts were invited to score, so 5

Table 3

Cluster word frequency analysis result of the riding scene.

Cluster	Review No.	Most Frequent Words (No. Of them)			
C2	1135	Comfortable (155)	Compact (113)	Lightweight (104)	Light & handy (68)
C3	32	Grab (7)	Riding (7)	Compact (4)	Touch (4)
C4	65	Pedal (9)	Sill step (8)	Pedal plate 8	Step (6)
C5	154	Sill step (61)	Pedal (54)	Back seat (12)	Space (7)
C6	96	Comfortable (49)	Seat (18)	Cozy (13)	Steady (5)

judgment matrix SIC was obtained. After calculating according to formulas 4 and 5, the scores of experts No.2 and No.4 were deleted, for the reason that the CR value of their judgment matrixes were 0.302 and 0.406, respectively. The CR value of the judgment matrixes from experts No.1, No.3 and No.5 were 0.087, 0.067, and 0.006, which were all less than 0.1. Then, according to the proposed method in Section 3.3, calculate the weight values given by the experts No.1, No.3 and No.5 and average the scores given by the three experts. The weight of three interaction interface criteria can be obtained. The three weights are : $wic_1 = 0.259, wic_2 = 0.663, wic_3 = 0.078$.

Based on the interactive emotion board of the riding scene obtained in Stage 2, the potential design points to improve the riding experience are clearly presented. The modification decisions of SEBSS in the riding scene were summarized by designers, as shown in Fig. 12. According to these modification decisions and the interactive emotion board, the

initial solution of smart electric bicycle is improved, and the solution after improvement is shown in Fig. 13. The foot pedal is combined with the bicycle body to give more space for the user's foot. The reminder mode is changed from handle vibration to voice reminder, and the shape and material of the seat are adjusted to make users feel comfortable.

79 target users gave satisfaction scores to the improved solution under three criteria. After calculating the average value of the satisfaction scores, the final total score was obtained:

$$S = A_1 \times wic_1 + A_2 \times wic_2 + A_3 \times wic_3$$

$$= 5.380 \times 0.259 + 5.228 \times 0.663 + 5.316 \times 0.078 = 5.274 \quad (7)$$

In this case, the design team wants to focus on improving the user experience, so they selected 5 as the standard. And the S value of SEBSS has met the standard, so the improved solution can be used in the subsequent embodiment and detail design stage.

Moreover, as mentioned in Section 3, designers also can use this method to analyze the touch points in all scenes. The case of SEBSS is used to demonstrate this. Firstly, electric bicycle has some additional services, e.g. door-to-door delivery, video-guided assembly, and maintenance. The touch points during the whole user experience process were summarized into 34 points, listed in Table 4.

Secondly, the target Kansei words and target design words for all touch points were selected. Then review samples were converted into matrix Z_E (3964×35 dimensional vector), and SOM cluster analysis was performed on it. The number of lattices was 13×13 . The clustering result was shown in Fig. 14, and it can be divided into 26 clusters. Word frequency analysis of the target design words and target Kansei words in the comments of clusters was conducted. And the clusters related to too many touch points (more than two) were regarded as invalid clusters,

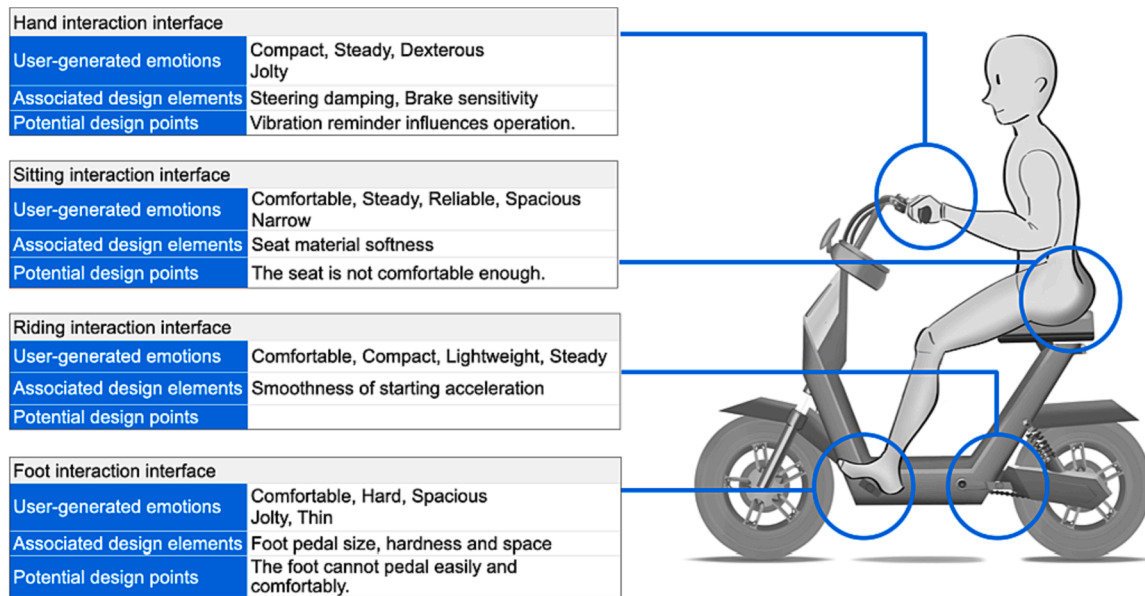
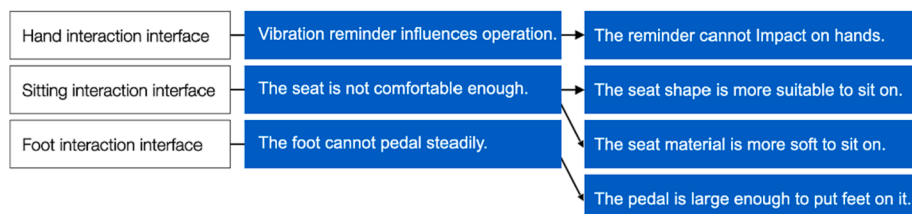
**Fig. 11.** Interactive emotion board of the riding scene.**Fig. 12.** The modification decisions of SEBSS.



Fig. 13. Solution after improvement.

Table 4
Touch points during the user journey of SEBSS.

Number	Touch point	Number	Touch point	Number	Touch point	Number	Touch point
1	Frame	10	Performance	19	Noise	27	Special user
2	Tool	11	Vibration	20	Vision	28	Cost performance
3	Seat	12	Speed	21	Somatosensory	29	Battery
4	Basket	13	Light	22	Workmanship	30	Logistics
5	Component	14	Assembly	23	Repair	31	Special scenario
6	Housing	15	Riding	24	Shape	32	Special weather
7	Pedal	16	Hand	25	Color	33	Special time
8	Wheel & Tyre	17	Foot	26	Defect	34	Special terrain
9	Anti-theft	18	Sitting				

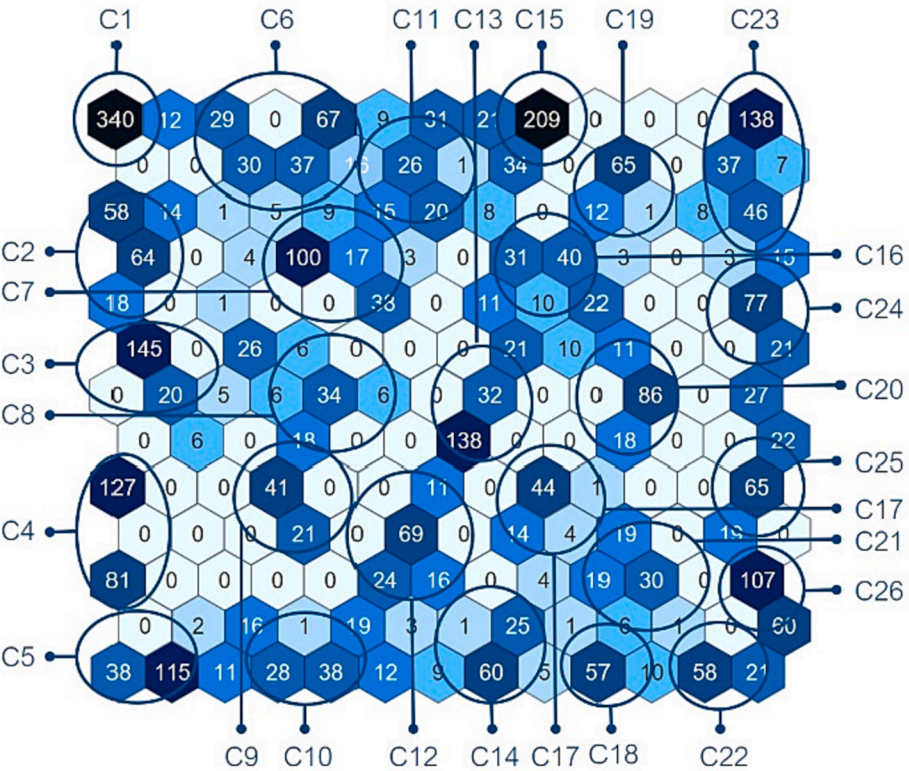


Fig. 14. SOM cluster analysis result of SEBSS.

including clusters 6, 18, 21, 22, 23, 24, 25, and 26. The remaining clusters were aligned to touch points frequently appearing, as shown in Table 5. Then, the user-generated emotions and associated design

elements of each touch point were determined from the high-frequency words and their comments under each touch point.

Finally, the potential design points were obtained through analysis of

Table 5

Cluster assigned to touch points.

Touch Point No.	Touch Point	Clusters
14	Assembly	C7 C14
22	Workmanship	C4 C5 C19
23	Repair	C11
24	Shape	C2 C17
27	Special user	C3 C8 C16
28	Cost performance	C1 C4 C5 C10 C12 C14
29	Battery	C7 C8 C9 C12 C13 C17 C20
30	Logistics	C11 C15 C16 C19 C20

the clusters assigned to each touch point, which need to be improved in the conceptual design solution of SEBSS. In this case, three touch points were considered to have the potential for improvement. Then the interactive emotion board was completed, as shown in Fig. 15.

5. Discussion

79 users were invited to give the satisfaction scores of two solutions. One solution was the initial solution of SEBSS (Fig. 7), and the other was the solution after improvement (Fig. 13). Compare the user satisfaction degree scores of these two solutions below. As shown in Fig. 16, the satisfaction scores of the three interaction interfaces are mainly compared. The riding interaction interface has not been modified, so it is not involved in the comparison. Firstly, compare the number of users who are dissatisfied with the solutions, i.e. the total number of users who give 1/2/3 point. The dissatisfied user number of the hand interaction interface was reduced from 15 of the initial solution to 10 of the modified solution. The number is reduced from 18 to 12 for the sitting interface and reduced from 14 to 10 for the foot interface. Secondly, compare the average value of user satisfaction degree scores to two solutions. The average satisfaction scores of the hand interaction interface are increased from 5.127 for the initial solution to 5.380 for the modified solution. The number increased from 4.861 to 5.228 for the sitting interface and increased from 5.051 to 5.316 for the foot interface. One can find that, after using the proposed method to improve the solution, the dissatisfied user number of the three interfaces is all reduced, and the average satisfaction scores of the three interfaces are all increased. It has been proved that the proposed method can improve user satisfaction degree in the conceptual design stage.

Meanwhile, the proposed method can leverage online review data to obtain information to assist in the design process. This method emphasizes that in the conceptual design stage, useful information can be extracted from the user's feedback on other similar traditional products

so as to improve the Smart PSS conceptual design solution. Traditional conceptual design methods mainly depend on the subjective thinking of designers. In the proposed method, the subjective thinking of the designers is still fully retained as the main creative. Meanwhile, the workforce of the development team can be saved by using the SOM clustering algorithm to analyze users' online comments.

6. Conclusion

Smart PSS, which emphasizes closed-loop design by utilizing user-generated information, has attracted ever-increasing attention among industry and academics. Most existing studies adopt a Smart PSS conceptual design method focusing on meeting the functional requirements but ignoring emotional requirements. Meanwhile, SCPs in Smart PSS mostly evolve from traditional product categories. However, massive information on traditional products has not been fully utilized, which can assist in the development of Smart PSS. Nevertheless, few studies have addressed this issue, let alone an algorithm-based method considering user-generated information and emotion for Smart PSS development. Therefore, this paper presents a novel SOM-based approach to support conceptual design improvement in the Smart PSS context with user emotions about traditional product categories to fill this gap. The main contribution of this paper can be concluded as follows: 1) The information on the traditional product category can be utilized to assist the development of Smart PSS through the proposed method. 2) User-generated emotions/feelings can be abstracted efficiently by utilizing the SOM algorithm in order to reduce the labor cost of the design team. 3) The Smart PSS conceptual design solution can be modified to improve user satisfaction during the conceptual design stage without real usage.

There are still some limitations in the research. Firstly, in Smart PSS, a large number of SCPs are involved. In the proposed method, the involved SCPs can only be analyzed separately. The comprehensive consideration of different SCPs in the system is not realized. Meanwhile, the proposed method is limited to the specific situation of the product-oriented Smart PSS and is not applicable to all Smart PSS projects. Furthermore, this study used only one case to verify the proposed method. More cases need to be used to prove the generalization of this method. Moreover, emotion characterization was not deeply studied in this research. Finally, in the analysis of review clustering, it is necessary to build the relationship between review samples and specific touch points. However, sometimes clustering clusters do not correspond to the touch points one by one, as in the case study, which will increase the analysis difficulty of developers. It is recommended that future research efforts can be dedicated to the following aspects: 1) Further improve the

	Assembly touch point	Logistics touch point	Special user touch point
User-generated emotions/feelings	Easy, smooth	Fast, reassuring	Safe, unworried
	Difficult, want to give up	Troublesome, upset	Worried, afraid, distracted
Associated design elements	Playing speed, playing mode	Inner packaging material, inner package shape, vibration absorption approach	Child seat position, fixing approach
Potential design points	The instruction video is too fast to follow.	The surface is easy to be scratched.	It's hard for a mother to ride an electric bicycle taking her child safely.

Fig. 15. Interactive emotion board of SEBSS.

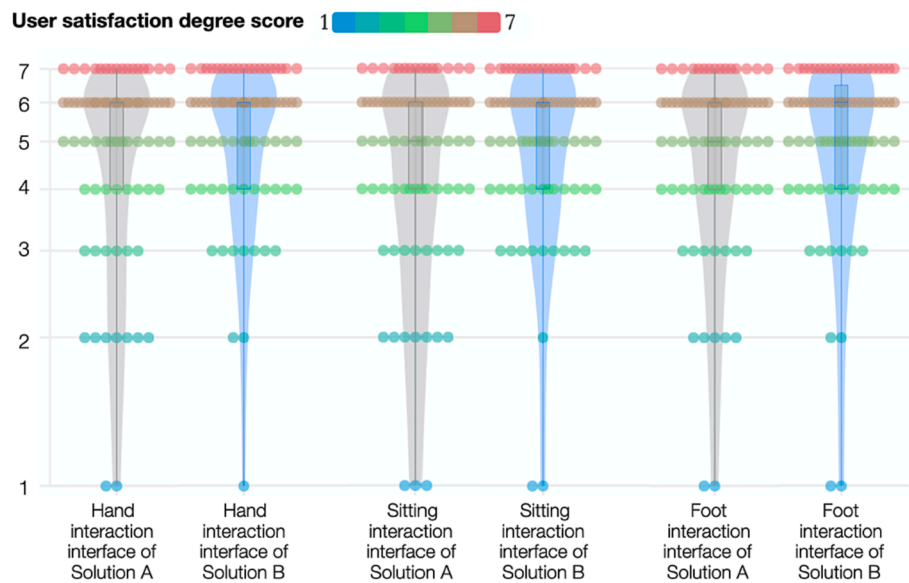


Fig. 16. Comparison between solutions before and after improvement.

automaticity and objectivity of this method. Reduce the involvement of experts and the subjective judgment of developers by introducing more effective algorithms and methods; 2) Enhance the ability of the proposed method to consider the involved SCPs and their relationship together so as to make it possible to comprehensively improve the initial solution of Smart PSS. The way to solve the design conflict in Smart PSS conceptual design can be proposed in future research. 3) In addition to information on traditional products, it should also be considered to develop the method of utilizing information on traditional service design to improve the Smart PSS solution. As an explorative study, it is desired that this research can provide a practical guide to the Smart PSS conceptual design process.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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